Further Mathematics GCE A Level 2021 - Paper 1

Duration: 1h30 | Cameroon GCE Board

GENERAL CERTIFICATE OF EDUCATION BOARD, General Certificate of Education Examination. 0775 FURTHER MATHEMATICS 1, ADVANCED LEVEL, JUNE 2021. Multiple choice question paper. Duration: one and a half hours. Answer ALL the 50 questions in this examination. All questions carry equal marks. Calculators are allowed. Each question has FOUR suggested answers: A, B, C and D. Decide which answer is correct, find the number of the question on the answer sheet and draw a horizontal line across the letter to join the square bracket for the answer you have chosen. Mark only one answer for each question (if more than one answer is marked, the question scores zero). Avoid spending too much time on any one question. Do all rough work in the booklet.

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1. The general solution of the differential equation $\cos x \dfrac{dy}{dx} - y \sin x = \sin x$ is

a) $y = -1 + k \sec x$
b) $y = 1 + k \sec x$
c) $y = (-1 + k) \sec x$
d) $y = (1 + k) \sec x$

2. $\dfrac{1}{(x+1)(x^2-1)}$, $x \neq \pm 1$, expressed in partial fractions for some constants p and q is

a) $\dfrac{p}{x+1} - \dfrac{\frac{1}{2}}{(x+1)^2} + \dfrac{\frac{1}{4}}{x-1}$
b) $\dfrac{\frac{1}{2}}{x+1} + \dfrac{px+q}{x^2-1}$
c) $\dfrac{p}{x+1} + \dfrac{\frac{1}{2}}{(x+1)^2} - \dfrac{\frac{1}{4}}{x-1}$
d) $\dfrac{p}{x+1} + \dfrac{\frac{1}{2}}{(x+1)^2} + \dfrac{\frac{1}{4}}{x-1}$

3. The value of c for which the function $f(x) = \begin{cases} x^2 + \dfrac{1}{2}c, & x \leq 0 \\ \dfrac{\sin x}{2x}, & x > 0 \end{cases}$ is continuous is

a) 0
b) $\dfrac{1}{2}$
c) 1
d) 2

4. The function $f(x)$ is such that $\dfrac{\cos x - 1}{x} + 3 \leq f(x) \leq \dfrac{\sin x}{x} + 2$, $\lim\limits_{x \to 0} f(x) =$

a) 0
b) 1
c) 2
d) 3

5. The polar equation of the hyperbola $xy = 4$ is

a) $r^2 = 8\,\text{cosec}\,\theta$
b) $r^2 = 2\,\text{cosec}\,\theta$
c) $r^2 = 8\,\text{cosec}\,2\theta$
d) $r^2 = 2\,\text{cosec}\,2\theta$

6. The equation of motion of a particle performing simple harmonic motion is given by $\dfrac{d^2y}{dx^2} + 18x = 0$. The period of the motion is

a) $\dfrac{\pi\sqrt{2}}{6}$
b) $\dfrac{\pi\sqrt{2}}{3}$
c) π
d) $\dfrac{\pi\sqrt{2}}{9}$

7. Given that $\mathbf{p}$ and $\mathbf{q}$ are two non-zero vectors such that $\mathbf{p} = \alpha\mathbf{q}$, where α is a real constant. Which one of the following is true?

a) $\mathbf{p} \cdot \mathbf{q} = 0$.
b) $\mathbf{p} \cdot \mathbf{q} = \alpha$
c) $\mathbf{p} \times \mathbf{q} = \mathbf{0}$.
d) $|\mathbf{p} \times \mathbf{q}| = \alpha$.

8. The structure $(\mathbb{Q}, \times)$ is NOT a group because $(\mathbb{Q}, \times)$

a) is not closed
b) is associative
c) does not have an identity element
d) does not have inverse elements

9. $x^2 \equiv -1 \pmod{25}$ is equivalent to

a) $x^2 \equiv 1 \pmod{25}$
b) $x^2 \equiv 24 \pmod{25}$
c) $x^2 \equiv 8 \pmod{25}$
d) $x^2 \equiv 5 \pmod{25}$

10. Given that $h\left(\dfrac{dy}{dx}\right)_n \approx y_{n+1} - y_n$ and $\dfrac{dy}{dx} = \dfrac{x}{y}, y \neq 0$, then $y_{n+1} \approx$

a) $h\dfrac{x_n}{y_n} - x_n$
b) $h\dfrac{x_n}{y_n} - y_n$

c) $h\dfrac{x_n}{y_n} + y_n$
d) $h\dfrac{x_n}{y_n} + x_n$

11. $\mathbf{k} \wedge (\mathbf{i} + \mathbf{j}) =$

a) $\mathbf{j} - \mathbf{i}$
b) $-\mathbf{j} + \mathbf{i}$
c) $-\mathbf{j} - \mathbf{i}$
d) $\mathbf{j} + \mathbf{i}$

12. If p and q are statements then $\sim (\sim p \Rightarrow \sim q) \equiv$

a) $p \wedge q$
b) $\sim p \wedge q$
c) $p \wedge \sim q$
d) $q \Rightarrow \sim p$

13. The value of x for which $\sinh^{-1}(x) = \ln 2$ is

a) $\dfrac{4}{3}$
b) $-\dfrac{1}{4}$
c) $\dfrac{5}{4}$
d) $\dfrac{3}{4}$

14. Given that $|z| = 5$, the minimum value of $|z - 3|$ is

a) 1
b) 2
c) 4
d) 8

15. For which of the following intervals is the Taylor series expansion of the function $f(x) = \ln(3 - 2x)$ valid?

a) $\left]-\dfrac{3}{2}, \dfrac{3}{2}\right]$
b) $\left[-\dfrac{3}{2}, \dfrac{3}{2}\right]$
c) $\left]-\dfrac{3}{2}, \dfrac{3}{2}\right[$
d) $\left[-\dfrac{3}{2}, \dfrac{3}{2}\right[$

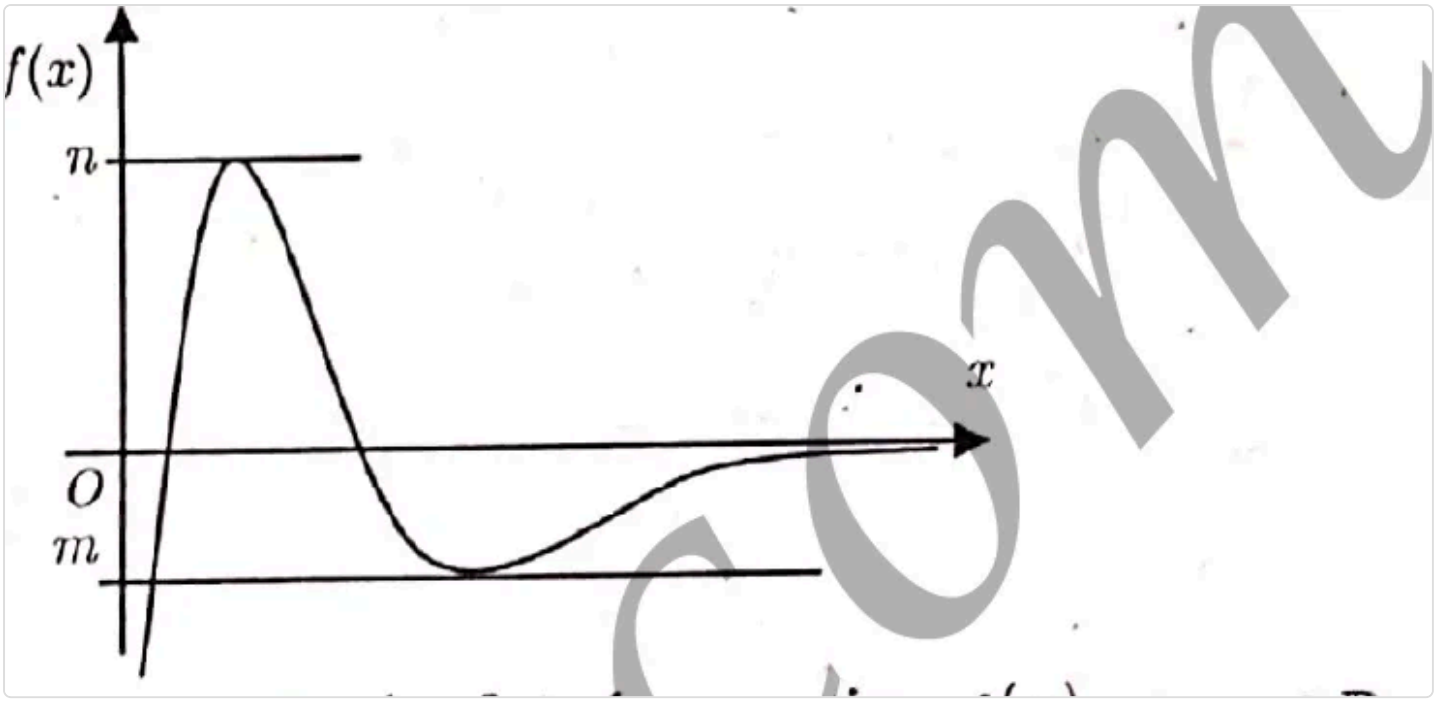
16. If $u_0 = 5$, $u_{n+1} = u_n - 3$, then which one of the following statements is true for U_n?

a) U_n is a geometric sequence
b) U_n is a decreasing sequence
c) U_n has positive terms
d) U_n is an increasing arithmetic sequence

17. $\int_{-2}^{2} |x|\, dx =$

a) 0
b) 1
c) 2
d) 4

18.



From the graph of f, the equation $f(x) = a$, $a \in \mathbb{R}$, has exactly two roots for

a) $a > 0$
b) $0 \leq a \leq n$
c) \$0 d) \$m

19. The root mean square value of $\cosh x$, for $1 \leq x \leq 3$ is

a) $\left[\frac{1}{2}\int \cosh^2 x\, dx\right]^{\frac{1}{2}}$
b) $\frac{1}{2}\left[\int_1^3 \cosh^2 x\, dx\right]^{\frac{1}{2}}$
c) $\frac{1}{3}\left[\int_1^3 \cosh^2 x\, dx\right]^{\frac{1}{2}}$
d) $\left[\frac{1}{2}\int_1^3 \cosh^2 x\, dx\right]^{\frac{1}{2}}$

20. The probability density function of a continuous random variable X is given by $f(x) = \begin{cases} a(1-x^2), & -\frac{1}{3} \leq x \leq \frac{1}{3} \\ 0, & \text{elsewhere} \end{cases}$ The value of a is

a) $\frac{81}{52}$
b) $\frac{162}{52}$
c) 1
d) $\frac{2}{3}$

21. The domain of the function $g(x) = \frac{x+1}{\sqrt{x^2-4}}$ is

a) $]-2, +\infty[$
b) $]-\infty, 2[$
c) $]-\infty, -2[\cup]2, +\infty[$
d) $]-2, 2[$

22. A sequence (u_n) is such that $\lim_{n\to\infty}(u_n) = 2$. Then (u_n) is ____ sequence

a) divergent
b) convergent
c) arithmetic
d) geometric

23. The image of the line $y = x$ under the transformation matrix $\begin{pmatrix} 1 & 0 \\ 2 & -3 \end{pmatrix}$ is the line

a) $y = x$
b) $y = 2x$
c) $y = -x$
d) $y = 1 - x$

24. $\int_0^1 \frac{1}{\sqrt{x^2+9}}\,dx =$

a) $\sinh^{-1}\left(\frac{1}{3}\right)$
b) $\cosh^{-1}\left(\frac{1}{3}\right)$
c) $\tanh^{-1}\left(\frac{1}{3}\right)$
d) $\cosh^{-1}(3)$

25. The work done when the point of application of the force $\mathbf{F}$ is displaced from $\mathbf{r}_1$ to $\mathbf{r}_2$ is

a) $\mathbf{r}_1 \times \mathbf{F}$
b) $\mathbf{r}_2 \times \mathbf{F}$
c) $(\mathbf{r}_1 - \mathbf{r}_2) \cdot \mathbf{F}$
d) $(\mathbf{r}_2 - \mathbf{r}_1) \cdot \mathbf{F}$

26. A smooth sphere is moving with velocity $(3\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$ when it hits a smooth wall parallel to the vector $\mathbf{j}$. If the coefficient of restitution between the sphere and the wall is $\frac{1}{3}$, then the velocity of the sphere after hitting the wall is

a) $\mathbf{i} + \mathbf{j}$
b) $\mathbf{i} - \mathbf{j}$

c) $-\mathbf{i}+\mathbf{j}$
d) $-\mathbf{i}-\mathbf{j}$

27. Given that $z, w \in \mathbb{C}$. Then the transformation $w = iz$ is

a) a rotation $\frac{\pi}{2}$ clockwise
b) a rotation $\frac{\pi}{2}$ anticlockwise
c) a rotation $\frac{\pi}{4}$ clockwise
d) a rotation $\frac{\pi}{4}$ anticlockwise

28. If $25\cosh x + 7\sinh x \equiv R\cosh(x+\lambda)$, then the value of R is

a) $\sqrt{674}$
b) 24
c) 18
d) 16

29. The moment of inertia of a rigid body of mass $3M$ about a given axis is $\frac{3Ma^2}{4}$. The radius of gyration about the axis is

a) $\frac{a}{2}$
b) $\frac{a}{4}$
c) $\frac{\sqrt{3}}{2}a$
d) $\frac{a^2}{4}$

30. The probability density function of a continuous random variable X is given by $f(x) = \begin{cases} a(1-x^2), & -1 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$ The value of a is

a) $\frac{4}{3}$
b) 1
c) $\frac{3}{4}$
d) $\frac{2}{3}$

31. The number, n, of bees in a large apiary reduces at a rate equal to thrice the number of bees present at time t. The differential equation that models the number of bees in this apiary is

a) $\frac{dn}{dt} = 3n$
b) $\frac{dn}{dt} = -3$
c) $\frac{dn}{dt} = 3$
d) $\frac{dn}{dt} = -3n$

32. The coordinates of the focus of the parabola $(y-4)^2 = 20(x+2)$ are

a) $(-2, 7)$
b) $(3, 4)$
c) $(-5, 4)$
d) $(-2, 1)$

33. One of the solutions of the congruent equation $x^2 - 3x + 4 \equiv 0 \pmod 7$ is

a) 3
b) 4
c) 5
d) 6

34. The integrating factor of the differential equation $\dfrac{dy}{dx} + 2y \tanh x = \cos x$ is

a) $\cosh x$
b) $\operatorname{sech} x$
c) $\cosh^2 x$
d) $\operatorname{sech}^2 x$

35. The negation of the statement $\sim P \wedge \sim Q$ is

a) $P \wedge Q$
b) $P \vee \sim Q$
c) $P \vee Q$
d) $\sim P \vee Q$

36. The work done by a force $\mathbf{F}$ in displacing a particle from a point $\mathbf{a}$ to the origin is

a) $-\mathbf{a} \cdot \mathbf{F}$
b) $\mathbf{a} \cdot \mathbf{F}$
c) $-\mathbf{a} \times \mathbf{F}$
d) $\mathbf{a} \times \mathbf{F}$

37. Given that $h\left(\dfrac{dy}{dx}\right)_n \approx y_{n+1} - y_n$ and $\dfrac{dy}{dx} = -(x + 2)$, then $y_{n+1} \approx$

a) $y_n + hx_n + 2$
b) $y_n + h(x_n + 2)$
c) $y_n + h(x_n - 2)$
d) $y_n - h(x_n + 2)$

38. The function $f(x) = \dfrac{(x^2 - 4)(x + 2)(x + 3)}{(x - 1)(x - 2)}$ has a removable discontinuity at $x =$

a) 0
b) 1
c) 2
d) 3

39. A particle moves along the curve $r = 1 - \sin\theta$ with constant angular velocity ω. The maximum speed occurs for $\theta =$

a) $\dfrac{\pi}{2}$
b) π
c) $\dfrac{3\pi}{2}$
d) 2π

40. For a parameter $t \in \mathbb{R}$, the image of the point $(t, 2t, 3t)$ under the transformation with matrix $M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ is

Options as printed: A) $(2t, 3t, t)$ B) $(t, 3t, 2t)$ C) $(t, 2t, 3t)$ D) $(2t, t, 3t)$

41. For what value(s) of n is $\displaystyle\int_0^1 x^n\,dx = \int_0^1 (1-x)^n\,dx$

a) $n \in \mathbb{R}$
b) $n = 2k,\ k \in \mathbb{Z}.$
c) $n = 2k+1,\ k \in \mathbb{Z}$
d) $n \notin \mathbb{R}$

42. $z = \cos\dfrac{\pi}{6} - i\sin\dfrac{\pi}{6}$, then $\left|z + \dfrac{1}{z}\right| =$

a) $\dfrac{\sqrt{3}}{2}$
b) 1
c) $\sqrt{3}$
d) $2\sqrt{3}$

43. A binary operation $\bullet$ is defined on $\mathbb{R}$ by $x \bullet y = x + y + 1$. The inverse of 4 is

a) -8
b) -6
c) -4
d) 0

44. The first three terms in the expansion of $e^{\frac{x}{3}}$ are

a) $1 + \dfrac{1}{3}x + \dfrac{1}{9}x^2$
b) $1 + \dfrac{1}{3}x + \dfrac{1}{18}x^2$
c) $1 + \dfrac{1}{3}x + \dfrac{2}{9}x^2$
d) $1 + \dfrac{1}{3}x + \dfrac{1}{6}x^2$

45. A sequence (u_n) is defined by $u_0 = 3$, $u_{n+1} = \dfrac{1}{6}u_n$. $\lim\limits_{n\to\infty}(u_n) =$

a) 0
b) $\dfrac{1}{6}$
c) 1
d) 3

46. The asymptote to the curve $y + 2 = 3^x$ is

a) $x = 0$
b) $y = -2$
c) $y = 2$
d) $x = 1$

47. The arc length of the curve $f(x) = \cosh x$ for $0 \leq x \leq \ln 5$ is

a) $\frac{12}{5}$
b) $\frac{5}{4}$
c) $\frac{5}{2}$
d) $\frac{5}{12}$

48. For any three non-singular matrices A, B and C, $A^{-1}\left(B^{-1}C\right)^{-1} =$

a) $A^{-1}C^{-1}B^{-1}$
b) $C^{-1}B^{-1}A^{-1}$
c) $A^{-1}C^{-1}B$
d) $B^{-1}A^{-1}C^{-1}$

49. The moment of momentum about O of a particle of mass m moving with velocity $\mathbf{v}$ through a point with position vector $\mathbf{r}$ is

a) $\mathbf{r} \times m\mathbf{v}$
b) $m\mathbf{v} \times \mathbf{r}$
c) $\mathbf{r}.m\mathbf{v}$
d) $m\mathbf{v}.\mathbf{r}$

50. Which one of the following functions is NOT differentiable in the interval $-1 \leq x \leq 1$

a) $y = |x - 1|$
b) $y = x^2$
c) $y = -x^2$
d) $y = |x| + 1$

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